

Gravitational-wave emission from multi-level scalar clouds: non-relativistic rates and line observables

Implementation note for `src/Numerics/gw_rates.jl` and `cloud_shifts.jl`

1 Setup and conventions

The cloud around a BH of mass M and spin a is a superposition of hydrogenic levels $j = (n, l, m)$,

$$\phi = \sum_j \sqrt{\frac{N_j}{2\mu}} (\psi_j e^{-i\omega_j t} + \text{c.c.}), \quad \int |\psi_j|^2 d^3x = 1, \quad (1)$$

with $\alpha = GM\mu$ and lengths in Bohr units $a_0 = 1/(\mu\alpha)$. The level set is $1 \leq m \leq l < n \leq N_{\text{max}}$ plus the truncation modes $(2l+1, 2l, 2m)$ with $2l+1 > N_{\text{max}}$. Occupations are $u_j = N_j/(GM^2)$ and time is $\tau = \mu t$. The spectrum is

$$\frac{\omega_{nlm}}{\mu} = 1 - \frac{\alpha^2}{2n^2} - \frac{\alpha^4}{8n^4} + \frac{\alpha^4}{n^4} \frac{2l-3n+1}{l+1/2} + \frac{2am\alpha^5}{n^3 l(l+1/2)(l+1)}. \quad (2)$$

GW emission enters the evolution equations as

$$\frac{du_a}{d\tau} \supset - \sum_b (1 + \delta_{ab}) \Gamma_{ab}^{\text{ann}} u_a u_b - \sum_{b: \omega_b < \omega_a} \Gamma_{a \rightarrow b}^{\text{tr}} u_a u_b + \sum_{b: \omega_b > \omega_a} \Gamma_{b \rightarrow a}^{\text{tr}} u_a u_b, \quad (3)$$

with dimensionless rates Γ . Annihilations $a + b \rightarrow g$ remove one quantum of each level (two for $a = b$). Transitions $a \rightarrow b + g$ are stimulated ($\propto u_a u_b$) and conserve the number of quanta. Gravitons leave the system, so the BH mass and spin are unaffected. Every rate is computed for every pair of levels.

2 Annihilations

The graviton wavelength ($|\mathbf{k}| = \omega_a + \omega_b \simeq 2\mu$) is short compared with the cloud ($ka_0 = 2/\alpha$), so no multipole expansion applies. We use the flat-space power for a harmonic source $T_{ij}(\mathbf{x})e^{-i\omega t} + \text{c.c.}$,

$$\frac{dP}{d\Omega} = \frac{G\omega^2}{\pi} \Lambda_{ij,kl}(\hat{\mathbf{k}}) \tilde{T}_{ij}^*(\mathbf{k}) \tilde{T}_{kl}(\mathbf{k}), \quad \tilde{T}_{ij}(\mathbf{k}) = \int d^3x T_{ij}(\mathbf{x}) e^{-i\mathbf{k}\cdot\mathbf{x}}, \quad (4)$$

with the transverse-traceless projector Λ . Only $\partial_i \phi \partial_j \phi$ survives the projection, since the remaining terms of T_{ij} are $\propto \delta_{ij}$. The source at frequency $\omega_a + \omega_b$ is

$$T_{ij} = \frac{\sqrt{N_a N_b}}{2\mu} (\partial_i \psi_a \partial_j \psi_b + \partial_j \psi_a \partial_i \psi_b) \quad (a \neq b), \quad T_{ij} = \frac{N_a}{2\mu} \partial_i \psi_a \partial_j \psi_a \quad (a = b). \quad (5)$$

The Fourier transform is evaluated exactly in four steps:

1. the gradient formula expresses $\nabla(R_{nl}Y_{lm})$ in terms of $g_{l\pm 1}(r)Y_{l\pm 1, m-q}$ along the spherical basis vectors;
2. products of two gradients reduce to Gaunt coefficients;

3. the plane-wave expansion of $e^{-i\mathbf{k}\cdot\mathbf{x}}$ is applied;
4. the radial integrals $\int_0^\infty r^p e^{-\beta r} j_J(kr) dr$ are evaluated in closed form. The single log-divergent term of the Hankel-function expansion contributes $\frac{(2J)!}{J!2^J} \arctan(k/\beta)$.

The remaining angular integral over $\hat{\mathbf{k}}$ has a polynomial integrand in $\cos\theta_k$ and is done by Gauss–Legendre quadrature, which is exact. The result is

$$\Gamma_{ab}^{\text{ann}}(\alpha) = \frac{\alpha^6}{2\pi} F_{ab}(2/\alpha), \quad F_{ab} = \int d\Omega \Lambda_{ij,kl} \tilde{S}_{ij}^* \tilde{S}_{kl}, \quad (6)$$

where $\tilde{S}$ is the source in Bohr units. We store the leading non-relativistic term $\Gamma_{ab}^{\text{ann}} = C_{ab} \alpha^{p_{ab}}$. It is extracted from F at $\alpha = 10^{-3}$ and 2×10^{-3} with a Richardson correction for the $O(\alpha)$ term; the result is stable to 10^{-5} for $l \leq 34$.

Generically $p_{ab} = 2(l_a + l_b) + 10$. For every pair with $l_a = l_b = 1$ the flat-space $O(\alpha^{14})$ term cancels and $p = 16$. With the BH potential included, the known result for $2p \times 2p$ is $dE/dt = \frac{484+9\pi^2}{23040} (M_c/M)^2 \alpha^{14}$, i.e. $C = (484 + 9\pi^2)/46080$. The large- k tail is set by the wavefunctions at the origin. The flat-space coefficients obey $C_{ab} = (1 + 3[a \neq b]) w_{n_a} w_{n_b} C_{211,211}$ with $w_n = |R_{n1}/r|_{r \rightarrow 0}^2 / |R_{21}/r|_{r \rightarrow 0}^2 = \frac{32}{3} \frac{n^2-1}{n^5}$; we verified this relation numerically. We therefore apply the same scaling to the α^{14} result for all $n p \times n' p$ pairs; this override can be switched off.

3 Transitions

For $a \rightarrow b$ the graviton wavelength is long ($\Delta\omega a_0 \sim \alpha$), so the leading multipoles are exact at leading order. The cross density and momentum density at $\Delta\omega = \omega_a - \omega_b$ are

$$\rho = \mu \sqrt{N_a N_b} \psi_a \psi_b^*, \quad \mathbf{p} = \frac{i}{2} \sqrt{N_a N_b} (\psi_a \nabla \psi_b^* - \psi_b^* \nabla \psi_a), \quad (7)$$

with mass moments $q_{Lm} = \int \rho r^L Y_{Lm}^*$ and current moments $J_{Lm} = L^{-1} \int (\mathbf{x} \times \mathbf{p}) \cdot \nabla (r^L Y_{Lm}^*)$. The flux (Thorne 1980; Blanchet), written in spherical moments, is

$$P = 2G \sum_{L \geq 2} \Delta\omega^{2L+2} \left[c_L \sum_m |q_{Lm}|^2 + s_L \sum_m |J_{Lm}|^2 \right], \quad (8)$$

$$c_L = \frac{4\pi(L+1)(L+2)}{(L-1)L[(2L+1)!!]^2}, \quad s_L = \frac{16\pi L(L+2)}{(L-1)(L+1)[(2L+1)!!]^2}, \quad (9)$$

where the factor 2 counts both $e^{\mp i\Delta\omega t}$ components. Parity allows mass moments for $(-1)^{l_a+l_b+L} = +1$ and current moments for -1 , with $\max(2, |m_a - m_b|, |l_a - l_b|) \leq L \leq l_a + l_b$. All allowed L are summed. Dividing by $\Delta\omega$ per transition gives

$$\Gamma_{a \rightarrow b}^{\text{tr}} = \sum_{L \text{ mass}} K_L \alpha^{2-2L} \delta^{2L+1} + \sum_{L \text{ current}} K_L \alpha^{4-2L} \delta^{2L+1}, \quad \delta = \Delta\omega/\mu. \quad (10)$$

The coefficients K_L are α -independent: radial moments (evaluated exactly in BigFloat) times angular integrals. δ is taken from Eq. (2), so both the rate and the emission direction follow $\alpha(t)$ and $a(t)$. The leading channel scales as α^{2L+4} (mass) or α^{2L+6} (current).

The flat-space kinetic-stress prescription (Arvanitaki & Dubovsky 2011 eq. 38; Arvanitaki et al. 2015 App. A) omits the Newtonian stress. It deviates from the quadrupole formula by a pair-dependent factor: 4 for all $\Delta l = 0$ transitions tested, and 1/121 for $433 \rightarrow 311$. The multipole formula follows from conservation of the total (matter plus Newtonian) stress-energy, and we use it.

4 Evolution

Tables of (C, p) and (K_L, L, type) are built per N_{max} by `src/scripts/build_gw_tables.jl`. Entries negligible at every $\alpha \leq 1$ are pruned ($C < 10^{-60}$; channels below 10^{-8} of the dominant one). At the start of a run, channels with $\Gamma \mu / \hbar < 10^{-10} \text{ yr}^{-1}$ at the initial (α, a) are dropped. This cut remains conservative because α only decreases as the BH loses mass. `solve_system` evaluates all remaining rates in every right-hand-side call at $\alpha(t) = GM(t)\mu$ and $a(t)$. The option `gw_model` selects `:nonrel` (this note), `:off`, or `:rel` (placeholder for relativistic rates).

5 Line observables

Each annihilation and transition channel is a line. From the evolved $u_j(t)$, $M(t)$ and $a(t)$:

$$f = \frac{\mu \omega_{\text{GW}}}{2\pi \hbar}, \quad \omega_{\text{GW}} = \omega_a + \omega_b \text{ or } \omega_a - \omega_b, \quad (11)$$

$$P = \Gamma u_a u_b G M_0^2 \frac{\mu}{\hbar} \times \hbar \omega_{\text{GW}}, \quad (12)$$

$$h_0 = \frac{\sqrt{8GP}}{c^{3/2} 2\pi f d}, \quad (13)$$

where M_0 is the initial BH mass (it fixes the normalisation of u). $h_0 = (A_+^2 + A_-^2)^{1/2}$ is the orientation-averaged amplitude at distance d , obtained from $P/(4\pi d^2) = \frac{c^3}{32\pi G} (2\pi f)^2 h_0^2$. `src/Nlevels_Runs/gw_lines_from_run.jl` writes $f(t)$, $P(t)$ and $h_0(t)$ for all lines above a relative peak-power floor.

6 Cloud-induced frequency shifts

The cloud's gravity and the quartic self-interaction $V \supset -\mu^2 \phi^4 / (24 f_a^2)$ shift the levels. At first order in the Schrödinger–Poisson / Gross–Pitaevskii system, keeping all terms resonant with level i ,

$$\frac{\delta \omega_i}{\mu} = -\alpha^3 \sum_j u_j [D_{ij} + (1 - \delta_{ij}) E_{ij}] - \frac{\alpha^5}{8} \left(\frac{M_{\text{pl}}}{f_a} \right)^2 \sum_j u_j (2 - \delta_{ij}) \chi_{ij}, \quad (14)$$

$$D_{ij} = \iint \frac{|\psi_i(\mathbf{x})|^2 |\psi_j(\mathbf{x}')|^2}{|\mathbf{x} - \mathbf{x}'|}, \quad E_{ij} = \iint \frac{\psi_i^* \psi_j(\mathbf{x}) \psi_j^* \psi_i(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|}, \quad \chi_{ij} = \int |\psi_i|^2 |\psi_j|^2. \quad (15)$$

E_{ij} is the exchange term. It arises because $\Phi[\psi_i \psi_j^*] \psi_j$ oscillates at ω_i . The integrals are evaluated exactly by multipole expansion, with the radial double integrals in closed form via incomplete gamma functions. The shifts are applied to $f(t)$ in post-processing, using every level occupied above 10^{-8} of the peak occupation.

7 Validation

quantity	this work	reference
$2p \times 2p$ flat space, all α	agrees to 10^{-12}	ABH15 Tab. VI (closed form)
$2p \times 2p$ flat, leading	$6.777 \times 10^{-5} \alpha^{16}$	AD11 eq. 44 ($9\pi/2^{26}$ form)
$3d \times 3d$	$2.360 \times 10^{-8} \alpha^{18}$	ABH15 Tab. VI (to 10^{-5})
$4f \times 4f$	$1.447 \times 10^{-11} \alpha^{22}$	ABH15 Tab. VI (to 10^{-5})
$3d \rightarrow 2p$	$5.24 \times 10^{-6} \alpha^{10}$	$5 \times 10^{-6} \alpha^{10}$ (BGLS21 Tab. IV)
transitions vs. quadrupole formula	agree to 10^{-13}	AD11 eq. 39, Q_{ij} by 3D quadrature
current moment	agrees to 10^{-15}	3D quadrature
$D_{1s,1s}$	5/8	hydrogen
D : 211–211, 211–322, 322–322	0.185, 0.109, 0.090	0.19, 0.11, 0.09 (BGLS21 G11–12)
$E_{211,322}$	0.015	omitted in BGLS21
quartic coefficients	1.17, 0.352, 0.144 ($\times 10^{-4}$)	1.2, 0.35, 0.14 (BGLS21 B15–16)

ABH15: Arvanitaki, Baryakhtar, Huang, arXiv:1411.2263. AD11: Arvanitaki, Dubovsky, arXiv:1004.3558. BGLS21: Baryakhtar, Galanis, Lasenby, Simon, arXiv:2011.11646. The tests are in `test/test_gw_rates.jl`.

8 Limitations

- All rates are leading order in α and use hydrogenic wavefunctions. The flat-space annihilation coefficients receive $O(1)$ corrections from the BH potential; these are included only for $l = 1$ pairs, via the scaling above. Relativistic (Teukolsky) rates are not implemented.
- The cloud shifts are not fed back into the evolution. Since $\Gamma^{\text{tr}} \propto \delta^{2L+1}$, the transition rates carry a relative error of order $(2L + 1) \delta\omega/\Delta\omega$. For $544 \rightarrow 322$ in a run with $u_{211} \approx 0.4$, $\delta\omega/\Delta\omega \approx 0.12$. For near-degenerate levels the shifts can reverse the emission direction.
- The strain is orientation-averaged; the angular patterns $dP/d\Omega$ are computed internally but not output.